Experiment-3.1

Aim:- Performing Static Analysis on a Suspicious File and System Behaviour Analysis

Mapped CO :- Understand the different types of Attacks, Vulnerabilities. **CO3**

Objective: - To understand performing static analysis on a suspicious file and analyzing system behavior using VirusTotal to detect potential malware.

Theory:- Static analysis on a suspicious file involves examining its characteristics without executing it to identify potential malicious traits. The process begins with gathering basic file information, such as its type, format, and cryptographic hashes (MD5, SHA-1, SHA-256), which can be compared against known malware databases like VirusTotal. Extracting readable text from the file using tools like strings or BinText helps identify suspicious elements such as URLs, IP addresses, API calls, and commands. Further analysis of the Portable Executable (PE) headers using PEStudio or CFF Explorer can reveal anomalies, including unusual section names, abnormal imports like VirtualAlloc or WriteProcessMemory, and inconsistencies in timestamps. Additionally, embedded resources can be inspected for hidden payloads or encryption methods, and packed files can be unpacked using tools like UPX. Checking the digital signature of an executable using sigcheck.exe helps determine its legitimacy, as unsigned or self-signed files can indicate potential threats.

System behavior analysis using VirusTotal provides deeper insights by allowing the submission of files, URLs, and domains for multi-engine scanning. When a file is uploaded, VirusTotal checks it against multiple antivirus databases and presents a detection score based on previous and real-time scans. The platform also offers dynamic analysis, where the file is executed in a sandbox environment to observe its behavior, including attempts to modify system files, make registry changes for persistence, establish network connections, or inject malicious code into other processes. VirusTotal Graph further aids in investigating file relationships, helping security analysts connect a file to known malware families or threat actor campaigns. Additionally, URL and domain reputation checks provide insights into phishing sites and malicious servers. For automation, VirusTotal's API allows security teams to integrate file and URL scanning into their workflows, improving threat intelligence and response capabilities. Combining static analysis with VirusTotal's behavioral insights enhances malware detection and investigation, making it a crucial approach in cybersecurity.

Step 1: File Collection & Environment Setup

1. Obtain a suspicious file (ensure it's in a controlled environment).
2. Move the file to a safe analysis environment (preferably a VM).
3. Ensure all necessary tools are installed.

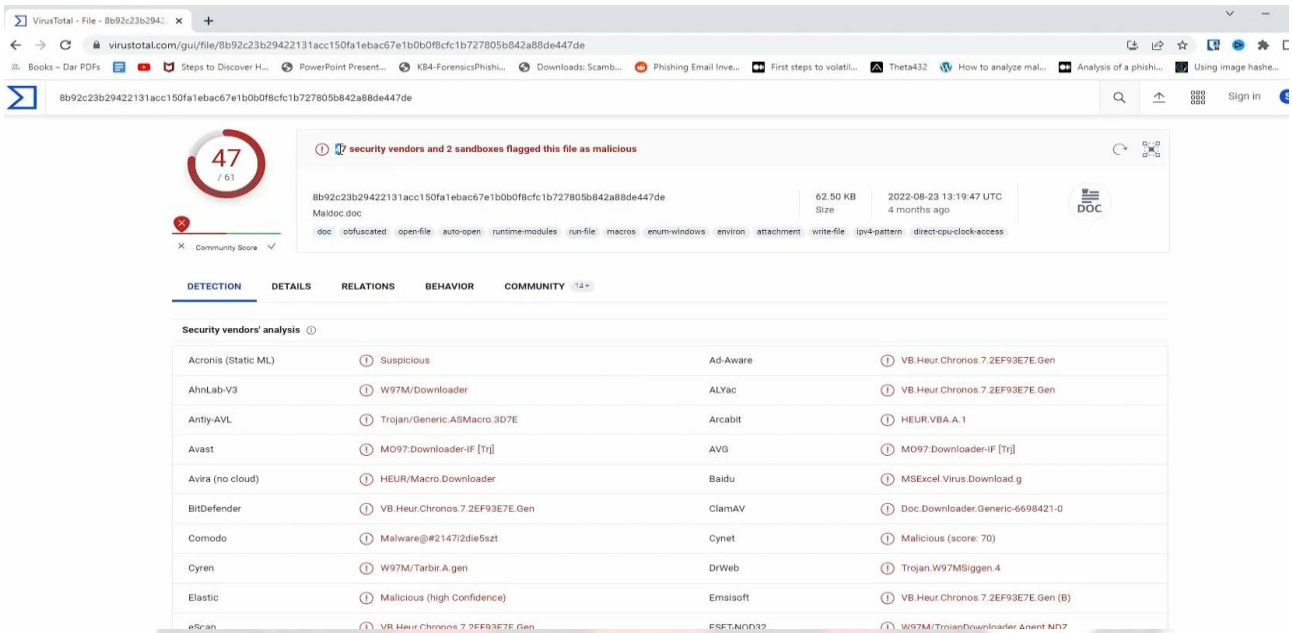
Step 2: Basic File Information Analysis

1. Check the file extension to identify its format (e.g., .exe, .dll, .doc, etc.).
2. Use the file command in Linux or right-click properties in Windows to get metadata.

3. Note the file size, creation date, and hashes (MD5, SHA-256) using:
    - Windows: certutil -hashfile <filename> MD5 / SHA256
    - Linux: md5sum <filename> / sha256sum <filename>

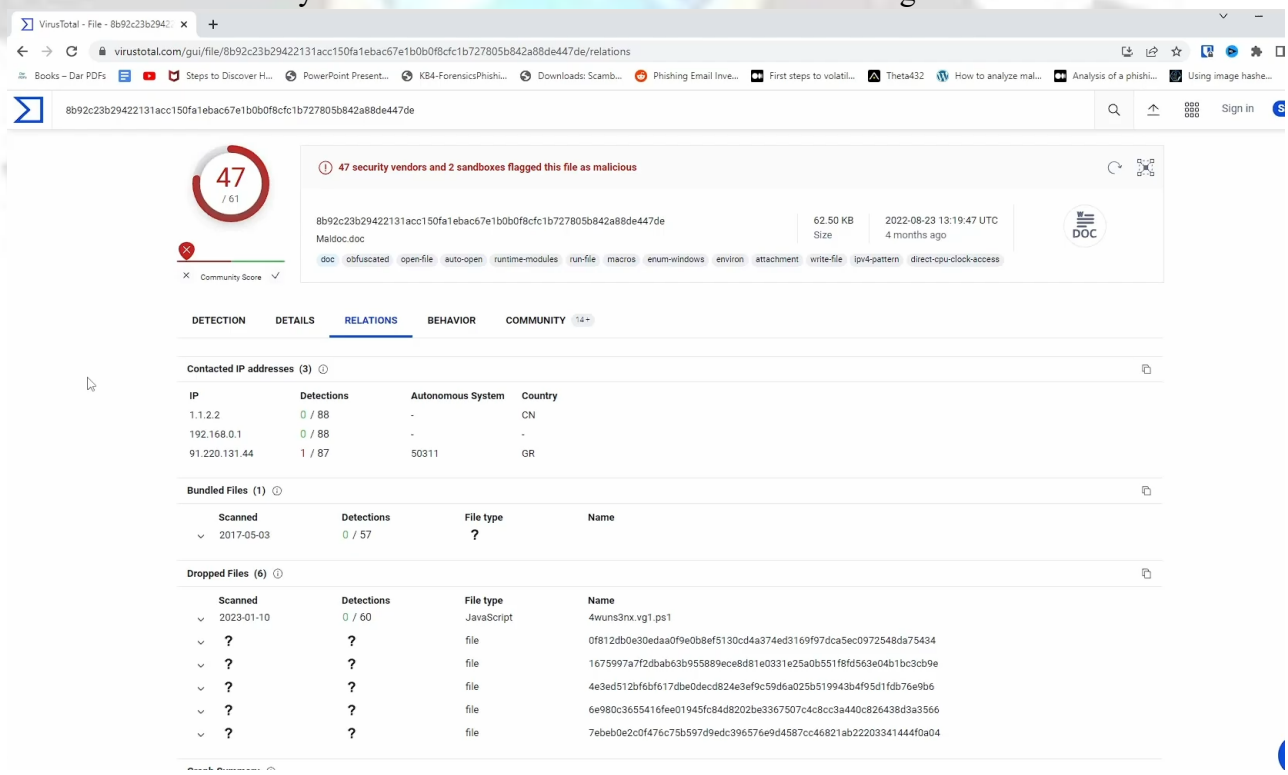
Step 3: Static Analysis

1. Open the file in a hex editor and inspect for suspicious headers or signatures.
2. Use **PE Studio** or **PEid** to analyze the file structure:
    - Identify compiler details and sections (e.g., .text, .data, .rdata).
    - Look for suspicious API calls and obfuscation techniques.
    - Check for digital signatures.
3. Run **Strings analysis**:
    - Use strings <filename> in Linux or tools like **Strings from Sysinternals**.
    - Look for suspicious URLs, IP addresses, registry keys, or encoded data.



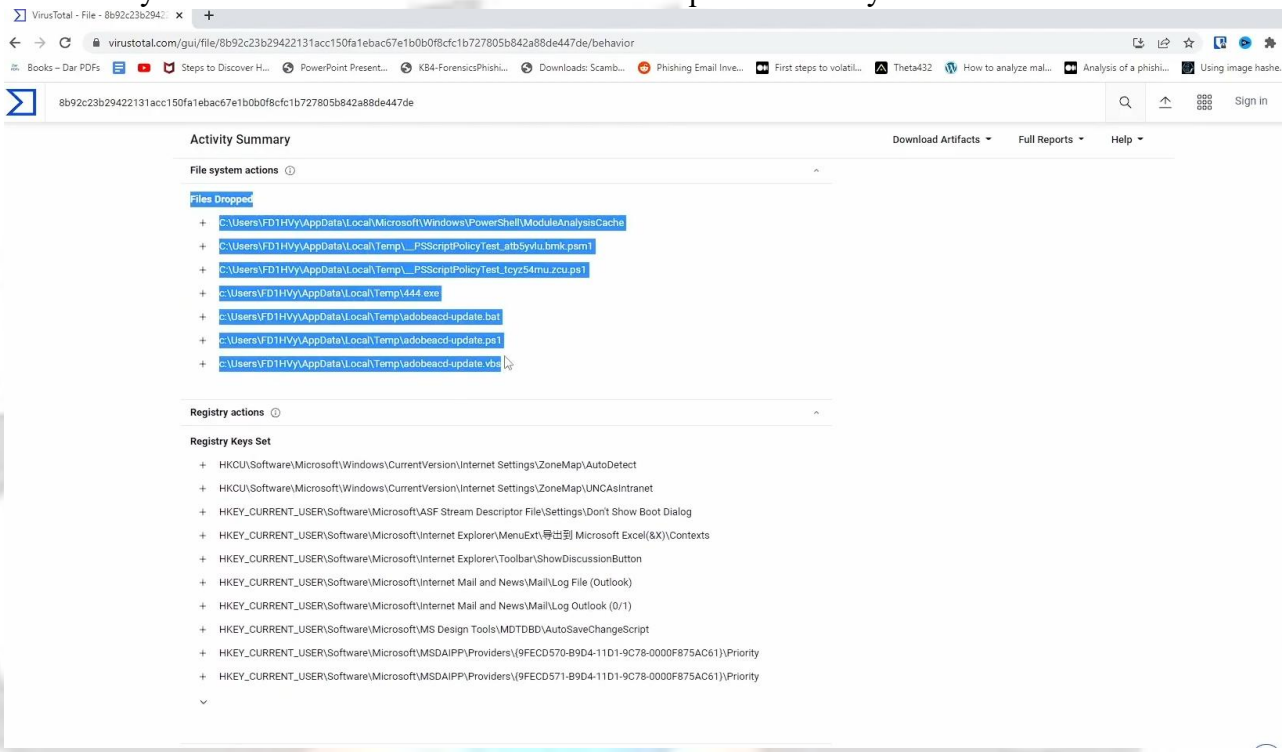
Step 4: VirusTotal Analysis

1. Go to VirusTotal.
2. Upload the suspicious file (or enter its hash if previously scanned).
3. Review scan results from multiple AV engines.
4. Analyze file behavior based on VirusTotal's behavioral insights.
5. Look at the Community tab for comments and additional threat intelligence.



Step 5: System Behavior Analysis (Optional)

1. If dynamic analysis is required:
   - Use tools like **Procmon, Wireshark, or Cuckoo Sandbox**.
   - Monitor file activity, registry changes, and network communications.
2. Review any abnormal behavior such as unauthorized processes or system modifications.



Step 6: Documentation & Reporting

1. Document all findings, including:
   - File metadata and hash values.
   - PE analysis results and suspicious indicators.
   - VirusTotal results and detections.
   - Any behavioral indicators.

Video Link: -

1. https://www.youtube.com/watch?v=p3AQjmEvApI

Viva Questions:

Q1. What is static analysis in the context of cybersecurity and malware detection?
Q2. How does static analysis differ from dynamic analysis?
Q3. What are the key advantages and disadvantages of static analysis?
Q4. What are some common tools used for static file analysis?